Scenario

Review the following scenario. Then complete the step-by-step instructions.

You are a newly hired cybersecurity analyst for an e-commerce company. The company stores information on a remote database server, since many of the employees work remotely from locations all around the world. Employees of the company regularly query, or request, data from the server to find potential customers. The database has been open to the public since the company's launch three years ago. As a cybersecurity professional, you recognize that keeping the database server open to the public is a serious vulnerability.

You are tasked with completing a vulnerability assessment of the situation to communicate the potential risks to decision makers at the company. You must create a written report that explains how the vulnerable server is a risk to business operations and how it can be secured.

Vulnerability Assessment Report

21st October 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. NIST SP 800-30 Rev. 1 is used to guide the risk analysis of the information system.

Purpose

Every data of the company like customer’s data, sales data, goods data, etc., are stored in the database server. If we don’t secure the database server and the security risk of database server is high it will lead to down of database server. This will impact the overall business which will lead to employees can’t access the server, customer can’t access the website until the database security risk issue is resolved.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Employee	Disrupt mission-critical operations.	*2*	*3*	*6*
Customer	Alter/Delete critical information	*1*	*3*	*3*
Hacker	Obtain sensitive information via exfiltration	*3*	*3*	*9*

Approach

Since the database server is open for public which is a vulnerability can cause many security events which can affect the organization. The threat source can be disgruntled employee, or customer, or hacker. Since the database server is open to public customer can knowlingly or unknowingly can delete information, liklihood of happening this is low because every customer will not have technical knowledge and if the event occurs the it can halt the business for some time, so the severity is high. Next any disgruntled employee can disrupt the business and he will have technical knowledge and access to the database, so the liklihood is medium and severity is high. And the last threat source is the hacker who will have all the technical knowledge and since database server is open for public, likelihood and severity of the threat is high.

Remediation Strategy

Since the database server is open we can implement some security controls which can mitigate the security threat. We can implement PKI which will encrypt the data while the data in use, rest or in transit. We can also implement authentication, authorization and accounting framework so that only authorized users can access the database and the log of every action can be recorded. We can also implement least privilege policy like Role based access control, depending on the role of every users we can set what they can access.